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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not required.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
The assay was repeated three times .
Number: The assay was repeated three times .
Data Information

Availability: Five-microliter aliquots were deposited on agar plates supplemented with Amp and Cm and plates with Amp , Cm and Sp .

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Recombinant DNA			
Biotinylated dsDNA fragments , used in the SPR analysis , were obtained by hybridization of two complementary single-stranded oligonucleotides (Metabion) ItT and ItB , pUCT and pUCB , as well as 1itT and 1itB .	pUCT		
	pUCB		
Plasmids (pSUMO-DBD and pSUMO-WH1WH2) were constructed by cloning the trfA gene fragments encoding the DBD and WH1WH2 domains into a pET SUMO vector .	pSUMO-DBD		
	pSUMO-WH1WH2		
	pET SUMO		Suggestion: RRID:Addgene_213623(link)
The constructions of pET SUMO-DBDP151S , pAT30trfA G254DS267LP151SE361K , pAT30trfA G254DS267LA171TP314S and pAT30trfA G254DS267LR103E , pAT30trfA G254DS267LT105E , pAT30trfA G254DS267LK165E , pAT30trfA G254DS267LP168E , pAT30trfA G254DS267LR169E , pAT30trfA G254DS267LN234E , pAT30trfA G254DS267LR236E , pAT30trfA G254DS267LK280E and pAT30trfA G254DS267LR347E plasmids were performed through site-directed mutagenesis based on the following templates: pET SUMO-DBD (using P151S-F and P151S-R oligonucleotides) , pAT30trfA G254DS267LP151S (using E361K-F and E361K-R oligonucleotides) , pAT30trfA G254DS267LA171T (using P314S-F and P314S-R oligonucleotides) and pAT30trfA G254DS267L (using R103E-F and R103E-R , T105E-F and T105E-R , K165E-F and K165E-R , P168E-F and P168E-R , R169E-F and R169E-R , N234E-F and N234E-R , R236E-F and R236E-R , K280E-F and K280E-R , R347E-F and R347E-R oligonucleotides) , respectively (Supplementary Table S2)	G254DS267LR347E		
The E . coli JM109 (pAT388) strain was transformed using pAT30 plasmids containing the trfA gene and grown on Luria Agar (LA)	pAT30		

Overnight cultures of JM109 (pAT388) , carrying the trfA wild-type or mutant gene , grown in Luria Broth (LB) medium with Amp and Cm , were diluted 1:100 in LB and grown for 2 h at 37degC .	pAT388		
The dsDNA was immobilized on the sensor surface to yield a final value of 60 RU for a DNA fragment containing a sequence of five iterons and for unspecific DNA of pUC19 and to 25 RU for DNA fragments containing a single iteron with the linker sequence.	pUC19		
Software and Algorithms			
Sequences were aligned separately using Clustal Omega v1.2.2 with default parameters (29) .	Clustal Omega		Suggestion: Clustal Omega (RRID:SCR_001591)(link)
Profile hidden Markov models (HMMs) were constructed with the hmmbuild program from the HMMER package (30) .	HMMER		
The HMM profile of the putative DBD was constructed based on sequence alignment of full-length TrfA homologs from the Pfam database (31)	Pfam		
One thousand ML searches were performed using RAxML v.	RAxML		
8.2.10 (32) with 100 rapid bootstrap replicates under the LG model of amino acid substitution and GAMMA model of rate heterogeneity with four discrete rate categories and the estimates proportion of invariable sites (LG + I + G) (33) , which was determined to be the best-fit model for all families by ProtTest v3.2 following the Akaike criterion (34) .	GAMMA		
	ProtTest		
After selecting 50 structures with least dihedral outliers (as judged by Molprobity (44)) , the thermodynamic effect of a number of selected amino acid substitutions on DNA affinity of TrfA was estimated for each of them using the mCSM webserver (50) .	Molprobity		
The SPR-derived experimental data together with data obtained with the mCSM webserver were used to select the best final model with the lowest-MSE (mean standard error).	mCSM		

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not a substitute for expert review. SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq